Discussion of "Stress Diffusion Experiment in Sand" by John H. Schmertmann

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The author presents results from surface plate loading tests that were used to investigate the effect that horizontal stress and the coefficient of lateral earth pressure, K, have on stress distribution and settlement. The author compares the results with a variety of solutions, and presents the practical implications of these findings. However, an explanation was not provided for these results, and the discussors believe that many of the author's findings may be explained as detailed below.

One of the primary findings reported by the author is that the ratio of the increase in vertical stress to the stress applied the plate ($\Delta\sigma_{vo}/q$) along the vertical axis under the center of the plate decreases as K increases. The concept of discrete stiff chains of particles that carry most of the load applied to the soil presented in Jardine et al. (1999) and Kuwano and Jardine (2002) would explain this result. They argue most particles are lightly loaded and the applied load is mainly focused into these stiff chains of sand particles. They also argue that anisotropically loaded samples have more discrete chains aligned along the major principal stress axis, whereas isotropically loaded samples have discrete chains that are more evenly distributed. Therefore, when the horizontal stress increases, as in the experiments performed by the author, more of these stiff chains would develop in the horizontal direction and carry a significant portion of the load. The increased number of horizontal stiff chains would carry more of the applied load, and the resulting increase in vertical stress under the plate would not be as great as in the tests with lower horizontal stresses.

Another result presented by the author is a decreasing amount of settlement with an increasing value of K for a given applied load. The Young's modulus for sand is a function of the density and the mean effective stress. Therefore, a larger value of K (keeping the vertical stress and load constant) would result in a higher mean effective stress and stiffness. This higher value of stiffness would result in less settlement, as found in the experiments by the author. If the model used in PLAXIS for the author's study employed a stiffness that is dependent on the mean effective stress (no details about the model used in PLAXIS were given by the author), the finite-element results should show the same trend in settlements as seen in the experiments.

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Closure to "Stress Diffusion Experiment in Sand" by John H. Schmertmann

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The writer thanks the discussers for their points raised, which seem to consist of two parts. The first considers a possible explanation for the experimental results in the paper involving "the concept of discrete stiff chains of particles." The second involves Fig. 6 and a question and a prediction of what the PLAXIS study showed about plate settlements with different values of K_i.

As to the first, the writer presented only experimental results and did not speculate about the sand-structure conditions that might have produced these results. Those interested in the "stiff chain" or any other concept now have some additional experimental data to consider.

As to the second, the PLAXIS model does include a stress dependent stiffness and the discussers made a correct prediction. The paper discusses the importance of lateral stress versus settlement and gives the Schmertmann (1972, 1985) and Handy (2001) references, as well as the Fig. 6 data. The PLAXIS results also showed a similar settlement dependence on lateral stress, and the writer has included these results in an additional figure. Fig. 1 of this closure superposes PLAXIS points on the relevant part of the experimental results shown in Fig. 6, and it includes both the I and II boundary conditions discussed in the finite-element study section. The writer also modeled both concentrated and uniform loading on the plate, which gave almost identical results. The settlement agreement shown in Fig. 1 seems to provide support for the reasonableness of the other PLAXIS results in Fig. 5.

Errata

The following corrections should be made to the original paper: Page 4, paragraph 2, line 10, change °F to °C. Page 9, line 20, remove "," after 0.3.

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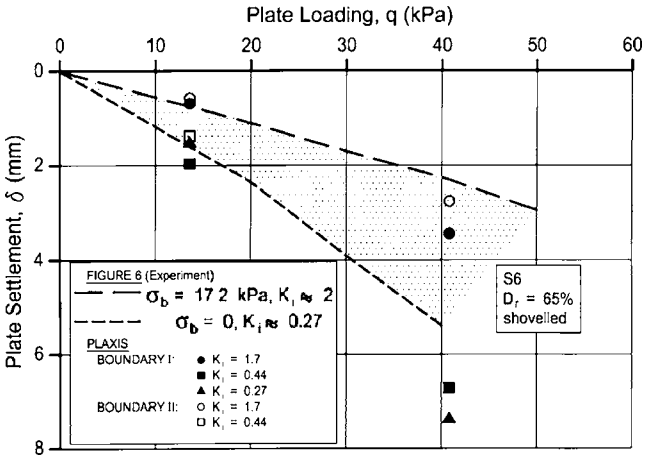

Fig. 1. Comparative settlements for PLAXIS

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Discussion of "Design Loading on Deeply Buried Box Culverts" by Kyungsik Kim and Chai H. Yoo

January 2005, Vol. 131, No. 1, pp. 20–27,
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It is recognized that the loads attracted to buried structures, from both overburden and surcharge loads, are governed by the characteristics of the soil, the geometry, and the stiffness of the structural components (Terzaghi 1943; Janssen 1895; Einstein and Shwartz 1979). In many instances the redistribution of "free-field" stresses as the result of the presence of a buried structure will result in a decrease in loading over the deflecting or yielding areas of the structure, and an increase over adjoining rigid or stationary parts (i.e., soil arching). This phenomenon can induce structural damage or collapse, or indeed can be utilized to optimize the efficiency of the structure. The discussers are conducting ongoing investigations of the behavior of scaled physical models of similar deeply buried box culverts using centrifuge modeling techniques (Stone et al. 1991; Stone and Newson 2002). Comparisons of the arching predictions made by the authors with the observed behavior of the discussers are encouraging and indicate that the two approaches create similar soil–structure interaction behavior. The discussers would like to add the following to the authors' contribution.

The authors have concentrated on the behavior of box culverts with uniform wall thicknesses and stiffness properties, and part of their research has investigated the effects of localized density variations of the fill overlying the culvert (the imperfect trench method). Similar soil–structure interaction can be observed if the thickness (or stiffness) of the structural elements of the culvert vary (i.e., that of the sidewalls and the top/bottom slabs). A series of scaled physical model tests have been conducted by the discussers, with the principal aim of investigating the loads attracted to box culverts due to variations in the thickness of the aforementioned structural elements. The 300-mm-long (L) model culverts were made from aluminum box section with an original external width dimension (B_c) of 101.6 mm and an initial wall thickness of 6.35 mm (see Fig. 1). Two opposing sides of this original section were machined to a wall thickness (t) of 2 mm; the other two sides remaining at their original thickness.

Miniature strain gauges were then attached to the culvert such that accurate estimates of the strains in the culvert sides could be monitored during subsequent calibration and centrifuge testing. The culverts were buried using air pluviation in a rectangular centrifuge strong box of dimensions 500-mm wide $\times$800-mm long$\times$600-mm high using Congleton sand. This material is a uniform fine sand with a d_{10}=100 μm, and maximum and minimum densities of 1.78 and 1.39 t/m^3, respectively. The relative density of the material was found to be Dr=30%. The thickness of sand overlying the culvert was $2\times H_c$ (i.e., twice the culvert thickness) and a further culvert thickness of sand lay below the structure. The strong box was placed on a large (7-m diameter) centrifuge and spun up to varying acceleration levels from 5 to 20 g and data from the strain gauges recorded continuously.

Fig. 2 shows the bending moment distribution deduced from the strain gauge output for the top side of the culvert at 5, 10, and 20 g, for Case A, where the sides are stiff (i.e., thickest) and the top and bottom slabs are flexible (i.e., thinnest). From these data it is evident that the bending moment distribution is parabolic. Best-fit curves have been matched to the data, and are also plotted in Fig. 2. The distributed load function (w_{meas}) is obtained from the second derivative of the bending moment function. For the parabolic bending moment distribution this corresponds to a uniformly distributed load, the magnitude of which is twice the second-order term coefficient. The uniformly distributed loads derived from the bending moment curves presented in Fig. 2 are

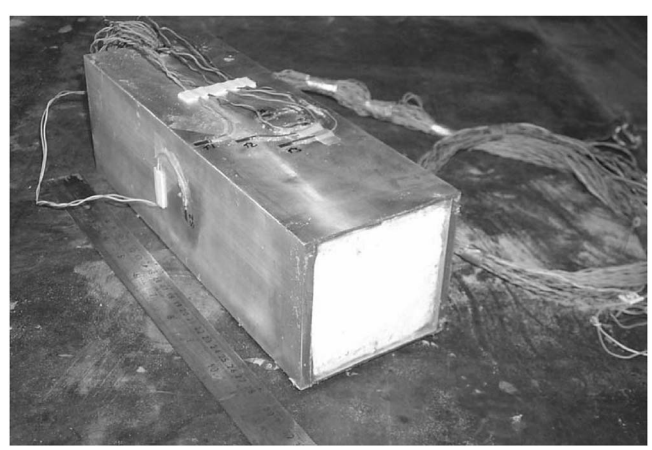

Fig. 1. Strain-gauged model box culvert prior to placing on sand bed

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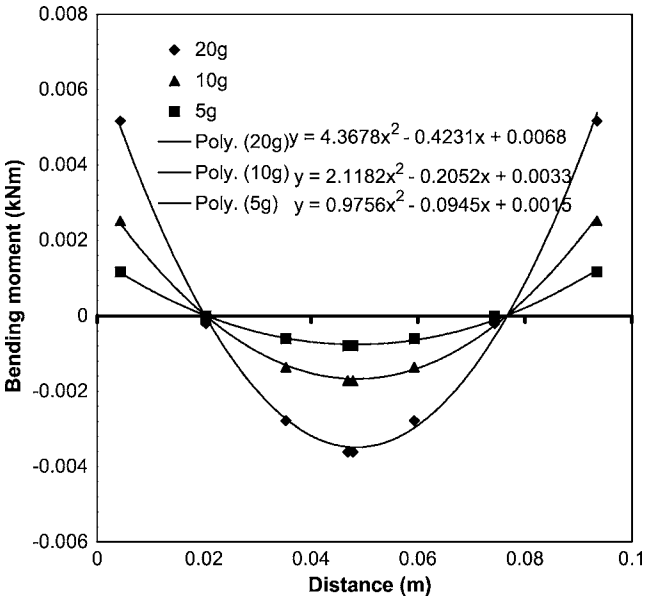

Fig. 2. Bending moments in top of culvert for Case A

shown in Table 1, together with the theoretical distributed load (w_{prism}) calculated from the weight of the assumed soil prism directly over the culvert.

It can be seen from the data presented in Table 1, that the load attracted to the top of the culvert is less than half of that calculated assuming a soil prism (with vertical sides) supported by the top slab of the culvert. Since the load attracted to the culvert, expressed as a proportion of the idealized soil prism load (w_{meas}/w_{prism}), remains constant over the range of applied g-levels, it can be deduced that the same volume of soil is involved in loading the top slab and that the load attracted to the top slab is independent of the top slab deflection.

Data from Case B, where the thin 2-mm sides of the culvert were orientated vertically to form the culvert sidewalls and the 6.35-mm thick sides formed the top and base of the culvert, are shown in Table 2. Again this shows a summary of the measured loads on the top slab and the comparison with the theoretical soil prism load. This indicates that as the g-level is increased, the load attracted to the culvert *increases* and eventually exceeds that corresponding to the assumed soil prism at about 10 g.

For Case A, the problem is not dissimilar to that of a deflecting trap door, since virtually all deformation is confined to the top and bottom faces of the culvert. The results for this case are thus consistent with active arching occurring over the deflecting top face of the culvert and the imperfect trench assumption. The degree of arching will depend on the geometry of the system (generally expressed in terms of the ratio of the overburden depth H to the width of the deflecting element—that is, the culvert slab width B_c) and the relative stiffness of the top slab. For the tests presented here the load attracted to the culvert top slab, for an H/B_c ratio of 2, is seen to be approximately 50% of the theoretical soil

Table 1. Summary of Loading Attracted to Top Side of Culvert for Test A

g-level	w_{prism} (kN/m)	w_{meas} (kN/m)	w_{meas}/w_{soil}
5	4.68	1.95	0.42
10	9.37	4.23	0.45
20	18.73	8.73	0.47

Table 2. Summary of Loading Attracted to Top Slab of Culvert for Test B

g-level	w_{prism} (kN/m)	w_{meas} (kN/m)	w_{meas}/w_{soil}
5	4.65	1.96	0.42
10	6.30	9.09	0.97
20	18.6	21.65	1.16

overburden load. This compares favorably with the numerical predictions of the authors who modeled a similar imperfect trench case with a soft zone above the box culvert. The results shown in Figs. 12 and 13 of the paper show reductions in load attracted to the top slab of the culvert between 40 and 60% for varying overburden ratios and soft zone stiffnesses.

The results presented in Table 2 for Case B are particularly interesting, and are presented together with the results for Case A in terms of the loading ratio (w_{meas}/w_{prism}) versus applied g-level (n) in Fig. 3. Initially, for Case B, the loading ratio is similar to that observed for test A, and suggests for the lightly loaded structure an active arching mechanism develops over the culvert top slab. This arching is not due to deflection of the top slab itself, but is likely to be the result of a combination of settlement of the whole culvert itself and a small settlement of the top slab as the result of the inward deflection of the culvert sidewalls. However, as the g-level increases the load attracted to the top slab increases, such that for applied g-levels greater than about 10 g, a loading ratio greater than one is recorded [i.e., the load attracted to the top slab (w_{meas}) is greater than the assumed soil prism loading (w_{prism})]. A possible explanation for this behavior is that with the culvert in this orientation, the inward movement of the sidewalls results in the formation of active Rankine zones in the soil immediately adjacent to them. Since the support to any arching mechanism over the culvert is provided by the soil immediately adjacent to the culvert top slab, as the active Rankine zones develop, this support is reduced, and any arching cannot be sustained. Furthermore, as the culvert sidewall deflections increase, the active soil zone increases toward the free surface resulting in an increase in load attracted to the culvert top slab. Thus the load attracted to the top slab eventually exceeds the assumed soil prism loading. This implies that the volume of soil involved in loading of the culvert top slab increases as the sidewall deflection increases, and consequently the load attracted to the culvert will be a function of the sidewall stiffness. However, it also implies that for low H/B_c ratios, then there will eventually be a maximum volume of soil involved in the interaction problem associated with fully developed active soil zones extending to the surface. This is tentatively

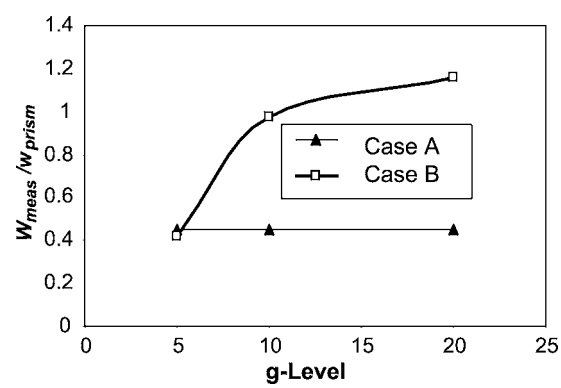

Fig. 3. Summary of top slab loading ratios, Cases A and B

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supported by the data shown in Fig. 3 suggesting that the load attracted to flexible sided culvert (Case B) is tending toward a limiting value. It is, however, not obvious if a similar observation would apply to the deeply buried culverts investigated by the authors.

Applying the observations of these culvert tests to the behavior of a uniform 2-mm-thick culvert, it would appear that the likely observed soil–structure interaction would be a combination of mechanisms associated with the relative deformation of the top slab and sidewalls. The reduction in load attracted to the top slab through active arching is likely to be partially offset by the increase in load attracted through the deflection of the sidewalls. Clearly the contribution of each mechanism will depend on the relative stiffness of the respective structural components.

The discussers plan to extend their program of centrifuge tests to investigate the behavior of deeply buried box culverts both for the perfect and imperfect trench conditions investigated numerically by the authors. It is hoped that further testing, including the use of uniform thickness culverts, will allow these observations to be quantified so that relationships can be developed to describe more fully the contribution of the interaction of individual structural elements to the overall soil–structure interaction problem over a range of burial depths.

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Closure to "Design Loading on Deeply Buried Box Culverts" by Kyungsik Kim and Chai H. Yoo

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The writers would like to thank the discussers for their interest in the paper and for sharing the results from their own experimental investigations. The discussers conducted experimental investigations of buried aluminum box culverts using the centrifuge modeling technique and pointed out that soil–structure interactions can in fact be measured.

Because both the magnitude and distribution of earth loads on buried structures depend on the relative stiffnesses of the buried structures and the soil, current design methods distinguish between a rigid (concrete) structure and a flexible (metal or thermoplastic) structure. AASHTO requires that metal structures are to be checked for strength limit state, including wall area, buckling, seam failure, flexibility limit for construction, flexure, etc. For the purpose of design, it is essential to distinguish between shallow and deep cover conditions depending on backfill heights. Under shallow cover conditions, live loads from vehicles are considered to be more significant than the load from soil. Under deep cover conditions, however, live load effects are not considered to be critical.

The writers wish to reiterate that loading on a rigid box culvert under deep cover conditions was treated in their paper. In the case of rigid structures under deep backfill, soil–structure interaction effects that arise due to the flexibility of the buried structures are not significant because the relative deflections of the buried structures are negligible. In this case, it is the differential settlement in the soil media (rather than the relative deflections of the buried structure) that causes considerable arching to develop, leading to the soil loads being redirected and redistributed and thus affecting the loads on the buried structures. In the case of flexible metal-plated culverts that were studied by the discussers, it can be expected that the considerable deflections of the top/bottom slab and sidewalls will affect the load exerted on the top slab depending on the relative stiffness of the structural components. It is the authors' hope that the theoretical investigation will be extended to the soil–structure interaction of flexible conduits with both deep and shallow backfills in the next phase of their ongoing research program.

Discussion of "Ultimate Lateral Resistance to Piles in Cohesionless Soils" by Linyang Zhang, Francisco Silva, and Ralph Grismala

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This discussion considers only one point, but an important one, of the original paper. It provides further support for the authors' choice of Prasad and Chari's 1999 lateral soil pressure and side shear distribution from Fig. 7 into Fig. 8.

In response to a consulting assignment from the Outdoor Advertising Association of America, Inc., in 1947, P.C. Rutledge (deceased), a former professor at Northwestern University and former partner in the engineering consulting firm of Mueser Rutledge in New York City, prepared the design nomograph in Fig. 1. In this nomograph he includes a design lateral pressure distribution, and the resultant horizontal forces Q_1 and Q_2, against

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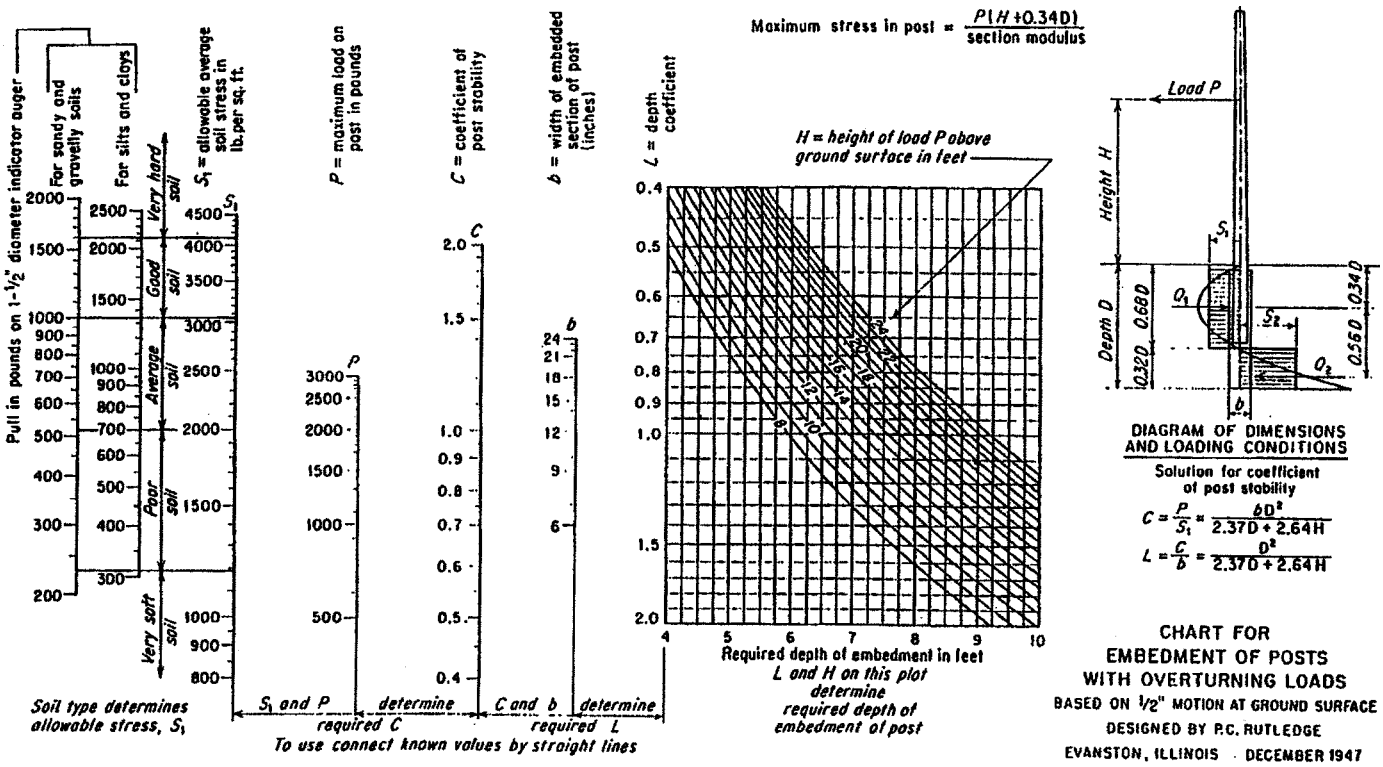

Fig. 1. Design nomograph

an embedded pole foundation resisting lateral wind loading against outdoor signs.

This discusser first became aware of this nomograph when Professor Rutledge distributed copies to his graduate students at Northwestern University in 1951. Patterson (1958) discusses its history, includes a copy, and credits his source.

Table 1 herein compares the Rutledge distribution with that by Prasad and Chari in Fig. 8. They match closely. I have occasionally used the Rutledge nomograph in practice and found it useful and have had no reason to question its accuracy. Considering this and the good results the authors obtained using the similar Prasad and Chari resistance distribution—58 years later!—I strongly support the authors' choice of this distribution for rigid piles.

Table 1. Comparison of Rutledge Design Nomography and Prasad and Chari (1999) Lateral Resistance Distributions

	Prasad and Chari	Rutledge
Rotation point	$0.72L$ (graphically)	$0.68L$
Line of upper resultant $Q1$	$0.38L$	$0.34L$
Line of lower resultant $Q2$	$0.91L$	$0.90L$
$Q1/Q2$	1.52	1.24 if $e/L=2$
		1.42 if $e/L=1$
		1.67 if $e/L=1/2$
		1.95 if $e/L=1/4$

Note: L=depth of pile embedment; Q used in Rutledge's notation; and e/L in P&C's notation.

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The writers would like to thank the discusser for comparing the lateral resistance distribution chosen in the paper for rigid piles with that prepared by Rutledge and for supporting the writers' choice. The writers must admit that they were not aware of the work by Rutledge when preparing the paper.

The discusser measured $a=0.72L$ from Fig. 8 and obtained the constant values of L_1/L, L_2/L, and Q_1/Q_2 in his Table 1 for the lateral resistance distribution of Prasad and Chari (1999), where a

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Table 1. Variation of a/L, L_1/L, L_2/L, and Q_1/Q_2 with e/L for the Prasad and Chari (1999) Distribution (P&C) and the P. C. Rutledge (1947) Distribution (PCR)

	a/L		L_1/L		L_2/L		Q_1/Q_2	
e/L	P&C	PCR	P&C	PCR	P&C	PCR	P&C	PCR
0.0	0.79		0.42		0.93		2.21	
0.25	0.76		0.40		0.92		1.85	1.95
0.3	0.75		0.40		0.92		1.79	
0.4	0.74		0.39		0.91		1.65	
0.5	0.73		0.39		0.91		1.59	1.67
0.6	0.72		0.39		0.91		1.53	
0.8	0.71		0.38		0.90		1.44	
1.0	0.70		0.37		0.90		1.38	1.42
1.2	0.69	0.68	0.37	0.34	0.90	0.90	1.34	
1.4	0.69		0.37		0.90		1.30	
1.6	0.68		0.36		0.89		1.27	
1.8	0.68		0.36		0.89		1.25	
2.0	0.68		0.36		0.89		1.23	1.24
2.2	0.67		0.36		0.89		1.21	
2.4	0.67		0.36		0.89		1.19	
2.6	0.67		0.36		0.89		1.18	
2.8	0.67		0.35		0.89		1.17	
3.0	0.66		0.35		0.89		1.16	

is the depth of the rotation point, L is the embedded length of the pile, Q_1 and Q_2 are the resultant force of the lateral resistance above and below the rotation point, and L_1 and L_2 are the depth of Q_1 and Q_2, respectively. However, the terms in the discusser's Table 1 for the Prasad and Chari (1999) distribution should not be constant but vary with e/L, where e is the eccentricity of loading. From Eq. 12a of the original paper, one can derive the following expression for a/L

$$a/L = [-(0.567 + 2.7e/L) + (5.307 + 7.29(e/L^2) + 10.541e/L)^{0.5}]/2.1996 \quad (1)$$

Table 1 shows the values of a/L from Eq. (1) and of the corresponding L_1/L, L_2/L, and Q_1/Q_2 at different values of e/L. Also shown in Table 1 are the corresponding values of a/L, L_1/L, L_2/L, and Q_1/Q_2 for the Rutledge distribution. Rutledge chose constant values for a/L, L_1/L, and L_2/L but values for Q_1/Q_2 varying with e/L. Considering the small range of a/L, L_1/L, and L_2/L, the calculations by using constant values provides a reasonable alternative, particularly in the late 1940s, prior to the availability of electronic calculators or computers.

Professor Schmertmann's observations and reference to Rutledge's 1947 work reinforce the writers' belief that, in this age of Internet searches and electronic databases, the geotechnical profession needs its senior members more than ever to impart perspective and avoid "reinventing the wheel."

Reference

Prasad, Y. V. S. N., and Chari, T. R. (1999). "Lateral capacity of model rigid piles in cohesionless soils." *Soils Found.*, 39(2), 21–29.

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